SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA1613 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

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QUESTIONS: 05 **Total Marks:50**

Annexure A

Pseudocode Writing Task (Algorithm Design)

Criteria	Problem Understanding *(2)*	Algorithm Design *(3)*	Use of Control Structures *(2)*	Pseudocode Structure *(1)*	Completeness *(2)*	***Total(10)***
Weightage	*20%*	30%	20%	10%	20%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I ___K.Nagapreetham(192525242)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Annexure A

Q1. Dataset: Online Shopping Transactions

Order ID **Products Purchased**

O1 Mobile, Earphones

O2 Mobile, Charger

O3 Earphones, Charger

O4 Mobile, Earphones, Charger O5

Mobile, Earphones

Question:

Design pseudocode for the **Apriori algorithm** to discover frequent product combinations from the online shopping dataset using a specified minimum support threshold. Clearly illustrate the candidate itemset generation and pruning process at each iteration.

Q2. Dataset: Library Book Borrowing Records

Borrow ID **Books Borrowed**

B1	Data Science, Python
B2	Python, Machine Learning
B3	Data Science, Machine Learning
B4	Data Science, Python, Machine Learning B5

Python, Machine Learning

Question:
Develop pseudocode for the **FP-Growth algorithm** to identify frequent borrowing patterns without generating candidate itemsets. Explain the steps involved in constructing the FP-Tree using the given borrowing records. *(BL4)*

Q3. Dataset: Frequently Purchased Services

Service Combination	Support Count
{Internet}	4
{Streaming}	4
{Cloud Storage}	3
{Internet, Streaming}	3
{Streaming, Cloud Storage}	2

Question:
Write pseudocode to generate **association rules** from the given frequent service combinations. Include the procedures for confidence calculation, rule generation, and filtering based on a minimum confidence threshold. *(BL3)*

Q4. Dataset: Student Course Enrollment

Student ID	Courses Enrolled
S1	Python, Database
S2	Python, AI
S3	Database, AI
S4	Python, Database, AI
S5	Python

Question:
Design pseudocode for **constraint-based frequent itemset mining** where only itemsets containing the course **"Python"**are considered during the mining process. Explain how the constraint affects candidate generation and pruning. *(BL4)*

Q5. Dataset: Hospital Treatment Hierarchy

Patient ID	**Treatments Received**
P1	Healthcare, Cardiology, ECG
P2	Healthcare, Orthopedics
P3	Healthcare, Cardiology
P4	Healthcare, Cardiology, ECG P5
	Healthcare, Orthopedics

Question:
Develop pseudocode for **multilevel association rule mining** using the treatment hierarchy provided in the dataset. Explain how rules are generated at different hierarchy levels and how concept hierarchies improve pattern discovery.

Q1. Apriori Algorithm for Online Shopping Transactions

The given dataset contains five online shopping transactions involving **Mobile, Earphones, and Charger**. The task is to design pseudocode for Apriori and show candidate generation and pruning.

Given Dataset

Order ID	**Products**
O1	Mobile, Earphones
O2	Mobile, Charger
O3	Earphones, Charger
O4	Mobile, Earphones, Charger
O5	Mobile, Earphones

Assumption

Since the PDF does not specify a numerical minimum-support threshold, for illustration we use:

$$Minimum\ Support = 60\%$$

Total transactions:

$$N = 5$$

Therefore,

$$Minimum\ Support\ Count = 0.60 \times 5 = 3$$

Step 1: Generate 1-Itemsets

Count each product:

Item	**Support Count**
Mobile	4
Earphones	3
Charger	3

All three satisfy the minimum support count of 3.

Therefore,

$$L_1 = \{Mobile, Earphones, Charger\}$$

Step 2: Generate Candidate 2-Itemsets

Possible candidates are:

$$C_2 = \{\{Mobile, Earphones\}, \{Mobile, Charger\}, \{Earphones, Charger\}\}$$

Their support counts are:

$$Support(Mobile, Earphones) = 3$$

$$Support(Mobile, Charger) = 2$$

$$Support(Earphones, Charger) = 2$$

After pruning:

$$L_2 = \{\{Mobile, Earphones\}\}$$

The other two itemsets are removed because their support count is less than 3.

Step 3: Generate 3-Itemset

The possible itemset is:

$$\{Mobile, Earphones, Charger\}$$

It occurs only in O4.

$$Support = 1$$

Since:

$$1 < 3$$

it is pruned.

Therefore, the algorithm terminates.

Apriori Pseudocode:

```
APRIORI(Database, MinimumSupport)

1. Scan the database.
2. Find support count of every 1-itemset.
3. Remove itemsets below MinimumSupport.
4. Store remaining itemsets as L1.
5. Set k = 2.

6. WHILE L(k-1) is not empty DO

      Generate candidate itemsets Ck
      from L(k-1).

      Prune candidates if any of their
      subsets are not frequent.

      Scan the database.

      Calculate support count of
      every remaining candidate.

      Remove candidates below
      MinimumSupport.

      Store remaining candidates as Lk.

      k = k + 1

   END WHILE

7. Return all frequent itemsets.
```

Result:

The main frequent combination obtained is:

$\{Mobile, Earphones\}$

Apriori principle: If an itemset is infrequent, its supersets cannot be frequent.

Q2. FP-Growth for Library Book Borrowing Records

The dataset contains borrowing records involving **Data Science, Python, and Machine Learning**. The question specifically asks for FP-Growth without generating candidate itemsets.

Given Dataset

Borrow ID	**Books**
B1	Data Science, Python
B2	Python, Machine Learning
B3	Data Science, Machine Learning
B4	Data Science, Python, Machine Learning
B5	Python, Machine Learning

Step 1: Calculate Frequency

Book	**Count**
Python	4
Machine Learning	4
Data Science	3

Assume: MINMUM SUPPORT COUNT = 3

All three books are frequent.

Step 2: Arrange Items

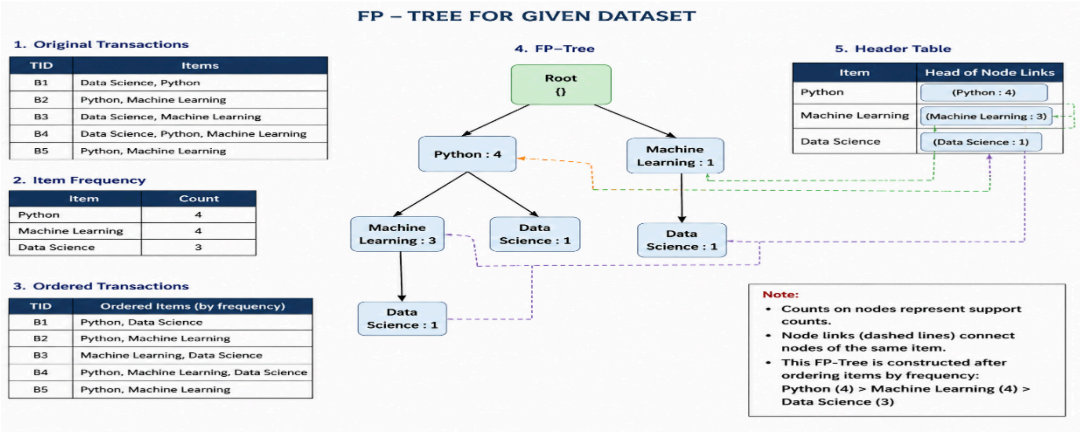
Items are arranged according to their frequency. A possible order is:

Transactions can then be reordered according to this frequency.

Step 3: Construct FP-Tree

The reordered transactions are inserted into a common tree.

A simplified representation is:



Common prefixes are stored only once, which reduces the size of the database representation.

FP-Growth Pseudocode:

```
FP-GROWTH(Database, MinimumSupport)

1. Scan the database.
2. Calculate frequency of every item.
3. Remove items below MinimumSupport.
4. Sort remaining items according to frequency.
5. Create an empty FP-Tree.
6. Insert each transaction into the FP-Tree.
7. Create a header table for frequent items.
8. Start mining from the FP-Tree.

9. FOR each item in the header table DO

       Construct its conditional pattern base.

       Construct the conditional FP-Tree.

       Recursively mine the conditional tree.

       Generate frequent patterns.

   END FOR

10. Return all frequent itemsets.
```

Important Point:

FP-Growth is different from Apriori because it **does not explicitly generate candidate itemsets**.

It uses:

Database→FP Tree→Conditional Pattern Base→Conditional FP Tree→Frequent Patterns

Q3. Association Rule Generation

The given frequent service combinations are **Internet, Streaming, Cloud Storage**, along with their support counts.

Given Data

Itemset	Support Count
{Internet}	4
{Streaming}	4
{Cloud Storage}	3
{Internet, Streaming}	3
{Streaming, Cloud Storage}	2

Formula for Confidence:

For a rule:

$$X \to Y$$

the confidence is:

$$Confidence(X \to Y) = \frac{Support(X \cup Y)}{Support(X)} \times 100$$

Rule 1: Internet → Streaming

$$Support(Internet) = 4$$

$$Support(Internet, Streaming) = 3$$

Therefore:

$$Confidence = \frac{3}{4} \times 100$$

$$= 75\%$$

Rule 2: Streaming → Internet

$$Confidence = \frac{3}{4} \times 100$$

$$= 75\%$$

Rule 3: Streaming → Cloud Storage

$$Confidence = \frac{2}{4} \times 100$$

$$= 50\%$$

Rule 4: Cloud Storage → Streaming

$$Confidence = \frac{2}{3} \times 100$$

$$= 66.67\%$$

Rule 4: Cloud Storage → Streaming

Assume Minimum Confidence = 70%

Rule	Confidence	Decision
Internet → Streaming	75%	Accepted
Streaming → Internet	75%	Accepted
Streaming → Cloud Storage	50%	Rejected
Cloud Storage → Streaming	66.67%	Rejected

Pseudocode:

```
GENERATE-RULES(FrequentItemsets, MinimumConfidence)

1. FOR each frequent itemset F DO

2.     Generate all non-empty proper
       subsets X of F.

3.     FOR each subset X DO

4.         Y = F - X

5.         Calculate:
           Confidence(X → Y)
           = Support(F) / Support(X)

6.         IF Confidence >= MinimumConfidence THEN
               Add X → Y to the rule set.
           ELSE
               Reject the rule.
           END IF

7.     END FOR

8. END FOR

9. Return all accepted rules.
```

Result:

For the assumed 70% threshold:

$$\boxed{Internet \rightarrow Streaming}$$

$$\boxed{Streaming \rightarrow Internet}$$

are accepted.

Q4. Constraint-Based Frequent Itemset Mining

The dataset contains student course enrollments. The important constraint is:

Only itemsets containing Python are considered

This constraint is explicitly specified in the question.

Given Dataset

Student	**Courses**
S1	Python, Database
S2	Python, AI
S3	Database, AI
S4	Python, Database, AI
S5	Python

\

Support Counts:

Python occurs in:

Upgrade

$$S1, S2, S4, S5$$

Therefore:

$$Support(Python) = 4$$

Python + Database occurs in:

$$S1, S4$$

Therefore:

$$Support(Python, Database) = 2$$

Python + AI occurs in:

$$S2, S4$$

Therefore:

$$Support(Python, AI) = 2$$

Python + Database + AI occurs only in S4:

$$Support(Python, Database, AI) = 1$$

Constraint-Based Pruning

Possible 2-itemsets:

$$\{Python, Database\}$$

$$\{Python, AI\}$$

$$\{Database, AI\}$$

But:

$$\{Database, AI\}$$

does not contain Python.

Therefore it is **pruned immediately**.

This reduces the search space.

Pseudocode:

```
CONSTRAINT-MINING(Database, MinimumSupport)

1. Generate candidate itemsets.

2. FOR each candidate C DO

       IF Python is not present in C THEN
            Prune C.
       ELSE
            Calculate support of C.

            IF Support(C) >= MinimumSupport THEN
                 Add C to frequent itemsets.
            ELSE
                 Prune C.
            END IF
       END IF

   END FOR

3. Generate larger candidates only if
   they contain Python.

4. Repeat until no further candidates exist.

5. Return frequent itemsets.
```

Advantage:

Constraint-based mining:

- Reduces candidate itemsets.
- Reduces support calculations.
- Saves processing time.
- Focuses only on relevant patterns.

Q5. Multilevel Association Rule Mining

The dataset contains a hierarchy of hospital treatments.

Given Dataset

Patient Treatments

P1	Healthcare, Cardiology, ECG
P2	Healthcare, Orthopedics
P3	Healthcare, Cardiology
P4	Healthcare, Cardiology, ECG
P5	Healthcare, Orthopedics

Treatment Hierarchy:

```
Healthcare
|
├── Cardiology
|     └── ECG
|
└── Orthopedics
```

This allows mining at different levels.

Level 1 – General Level

Healthcare appears in all 5 transactions:

$$Support(Healthcare) = \frac{5}{5} \times 100$$

$$= 100\%$$

Level 2 – Intermediate Level

Cardiology appears in:

$$P1, P3, P4$$

Therefore:

$$Support(Cardiology) = \frac{3}{5} \times 100 = 60\%$$

Orthopedics appears in:

$$P2, P5$$

Therefore:

$$Support(Orthopedics) = \frac{2}{5} \times 100 = 40\%$$

Level 3 – Specific Level

ECG appears in:

$$P1, P4$$

Therefore:

$$Support(ECG) = \frac{2}{5} \times 100 = 40\%$$

Example Rule

Consider:

$$Cardiology \rightarrow ECG$$

Cardiology occurs 3 times and both Cardiology and ECG occur together 2 times.

Therefore:

$$Confidence = \frac{2}{3} \times 100$$
$$= 66.67\%$$

Pseudocode:

```
MULTILEVEL-MINING(Database, Hierarchy)

1. Start from the highest level
   of the concept hierarchy.

2. Calculate support of items.

3. Remove items below MinimumSupport.

4. Generate frequent itemsets.

5. Generate association rules.

6. Calculate confidence.

7. Keep rules satisfying
   MinimumConfidence.

8. Move to the next lower level
   of the hierarchy.

9. Repeat Steps 2–8.

10. Stop at the lowest hierarchy level.

11. Return all interesting rules.
```

Importance of Concept Hierarchy

Concept hierarchies allow data to be analyzed from **general to specific**:

$$Healthcare \rightarrow Cardiology \rightarrow ECG$$

This helps discover:

- Broad patterns at higher levels.
- Detailed patterns at lower levels.
- Relationships between general and specific treatments.
- More meaningful patterns for decision-making.